

Expressions

Code generation for literals, terms and arithmetic; registers as a stack; and encoding an instruction, which is Gödel numbering made mechanical.

[← CC 04 · Symbols and Types](#)

THREE PROBLEMS

Where code goes. Where data goes. Which register.

CODE

Straight code generation: emit each instruction as soon as it is known, one after another, into `code_binary`; `code_size` grows by four per instruction. When a target address is not yet known, remember the spot and *fix it up* later. Next week.

DATA

Strings, big integers and global variables go into `data_binary`, in reverse: the first word is at offset -8 from `gp`, the next at -16 . The data segment is static allocation at compile time.

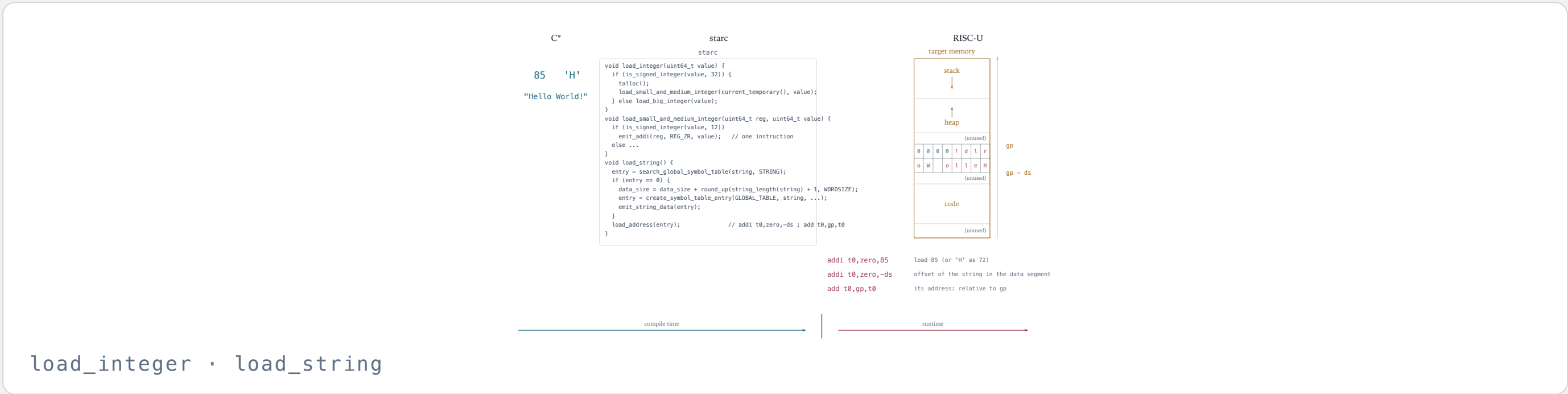
REGISTERS

Seven temporaries, `t0` to `t6`, allocated as a stack at compile time: `talloc`, `tfree`, `current_temporary`, `previous_temporary`. Knowing the two most recent is enough for every situation selfie meets.

A stack allocator at compile time, for registers used at runtime. The two timelines again — and the assertions on `allocated_temporaries` in the comments are how the compiler keeps itself from leaking registers.

LITERALS

Emitting a literal: one addi, or a word in the data segment and its address.



A small integer fits in the 12-bit immediate of one `addi t0,zero,85`. A medium one needs a `lui` and an `addi`. A big one goes into the data segment like a string and is loaded from there.

A string is emitted into the data segment, entered into the symbol table so that it is reused if it occurs again, and its *address* is what the code loads: `addi t0,zero,-ds` then `add t0,gp,t0`. The value of a string literal is a pointer.

TERMS

Infix in, postfix out: both factors in temporaries, then one instruction.

C*

x * 7

starc

```
uint64_t compile_term() {
    ltype = compile_factor();
    // assert: allocated temporaries == n + 1
    while (is_mult_or_div_or_rem()) {
        operator_symbol = symbol;
        get_symbol();
        rtype = compile_factor();
        // assert: allocated temporaries == n + 2
        if (ltype != rtype) type_warning(ltype, rtype);
        if (operator_symbol == SYM_ASTERISK)
            emit_mul(previous_temporary(), previous_temporary(), current_temporary());
        else if (operator_symbol == SYM_DIVISION)
            emit_divu(previous_temporary(), previous_temporary(), current_temporary());
        else if (operator_symbol == SYM_REMAINDER)
            emit_remu(previous_temporary(), previous_temporary(), current_temporary());
        tfree();
    }
    return ltype;
}
```

compiler memory

stack

↑

UINT64_T

SYM_MUL

UINT64_T

↑

heap

(unused)

code

(unused)

ltype

operator

rtype

global symbol table

42

x

gp

stack

↑

heap

(unused)

code

(unused)

ld t0, -ds(gp)

addi t1, zero, 7

mul t0, t0, t1

term = factor { ("*" | "/" | "%") factor } .

factor = [cast] ["-"] ["sizeof" "(" type ")"] [literal | identifier | call | "(" expression ")"] .

compile time

runtime

compile_term · x * 7

compile_term calls compile_factor, which leaves the value in a fresh temporary. Then while the symbol is *, / or %: remember the operator, get the next symbol, call compile_factor again, emit mul, divu or remu on the previous and current temporaries — and free one.

The infix operator is remembered across the second operand and emitted after it. That is the translation from the notation for humans to the notation for the machine, and the register stack is what makes it one instruction.

ARITHMETIC AND COMPARISON

Pointer arithmetic is **scaled**. A comparison is a **subtraction**.

C*

uint64_t* x;
x + 7
uint64_t* x;

starc

uint64_t compile_arithmetic() {
 ltype = compile_term();
 while (is_plus_or_minus()) {
 operator_symbol = symbol;
 get_symbol();
 rtype = compile_term();
 if (operator_symbol == SYM_PLUS) {
 if (ltype == UINT64STAR_T) {
 if (rtype == UINT64_T) {
 // pointer arithmetic: right term * SIZEOFUINT
 emit_multiply_by(current_temporary(), SIZEOFUINT);
 } else if (rtype == UINT64STAR_T) { ... }
 emit_add(previous_temporary(), previous_temporary(), current_temporary());
 } else if (operator_symbol == SYM_MINUS) {
 ...
 emit_sub(previous_temporary(), previous_temporary(), current_temporary());
 }
 tfree(1);
 }
 return ltype;
 }
}

compiler memory

stack
↓
UINT64STAR_T
SYM_PLUS
UINT64_T
↑
heap
(unused)
...
code
(unused)

ltype
operator
rtype

global symbol table

42
(unused)

code
(unused)

ld t0,-ds(gp)
addi t1,zero,7
addi t2,zero,8
mul t1,t1,t2
add t0,t0,t1

RISC-U

target memory

stack
↓
heap
(unused)
42
(unused)
code
(unused)

gp
x

expression = arithmetic [("==" | "!=" | "<" | ">" | "<=" | ">=") arithmetic] .
arithmetic = term { ("+" | "-") term } .

compile time
runtime

x + 7 with x a pointer: multiply by 8 first

C*

x == 7

starc

uint64_t compile_expression() {
 ltype = compile_arithmetic();
 if (is_comparison()) {
 operator_symbol = symbol;
 get_symbol();
 rtype = compile_arithmetic();
 if (ltype != rtype) type_warning(ltype, rtype);
 // for lack of boolean type
 ltype = UINT64_T;
 if (operator_symbol == SYM_EQUALITY) {
 // a == b iff unsigned b - a < 1
 emit_sub(previous_temporary(), current_temporary(), previous_temporary());
 emit_addi(current_temporary(), REG_ZR, 1);
 emit_sltu(previous_temporary(), previous_temporary(), current_temporary());
 tfree(1);
 } else if (operator_symbol == SYM_NOTEQ) { ... }
 else if (operator_symbol == SYM_LT) {
 // a < b
 emit_sltu(previous_temporary(), previous_temporary(), current_temporary());
 tfree(1);
 } else ...
 return ltype;
 }
}

compiler memory

stack
↓
UINT64_T
SYM_EQUALITY
UINT64_T
↑
heap
(unused)
...
code
(unused)

ltype
operator
rtype

global symbol table

42
(unused)

code
(unused)

ld t0,-ds(gp)
addi t1,zero,7
sub t0,t1,t0
addi t1,zero,1
sltu t0,t0,t1

RISC-U

target memory

stack
↓
heap
(unused)
42
(unused)
code
(unused)

gp
x

expression = arithmetic [("==" | "!=" | "<" | ">" | "<=" | ">=") arithmetic] .

compile time
runtime

x == 7 as unsigned 7 - x < 1

An instruction is a number. Making it one is Gödel numbering, by hand.

```
uint64_t encode_i_format(uint64_t immediate, uint64_t r
                        uint64_t funct3, uint64_t rd,
// assert: -2^11 <= immediate < 2^11
check_immediate_range(immediate, 12);
immediate = sign_shrink(immediate, 12);
return left_shift(left_shift(left_shift(left_shift(
    immediate, 5) + rs1, 3) + funct3, 5) + rd, 7) + opc
}
void emit_addi(uint64_t rd, uint64_t rs1, uint64_t imme
    emit_instruction(encode_i_format(immediate, rs1, F3_A
}
```

Twelve bits of immediate, five of source register, three of function, five of destination, seven of opcode: shift and add, and `addi t1, zero, 7` is `0x00700313`. The emulator's decoder does the same arithmetic backwards, and both are in one file, so the encoding is defined once.

This is the moment the compiler turns notation into a number that a machine can compute with. The introduction class called it Gödelisierung and made a theorem of it; here it is thirteen lines and a shift.

And the binary is a file that real hardware runs.

```
$ ./selfie -c selfie.c -o selfie.m && od -A d -t x1 sel  
00000000 7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00  
$ spike pk selfie.m  
selfie.m { -c { source } | -o binary | ... }
```

`-o` writes an ELF file: a header that says where code and data go, then `code_binary`, then `data_binary`. The same bytes load into mipster, into spike, and onto a RISC-V board.

Everything on this deck happened at compile time and produced a number. Everything that number does happens at runtime, on a machine that has never seen the source. That separation is the whole architecture of a compiler.

ASSIGNMENT

bitwise-shift-execution: sll and srl, in the ISA, the emulator, and the code generator.

- 1 Specify. Extend `riscu.md` with `sll rd,rs1,rs2` and `srl rd,rs1,rs2`: assembly, encoding as R-format instructions, and semantics in the style of the other lines.
- 2 Encode and decode. An `emit_sll` and `emit_srl` using `encode_r_format`, and the matching cases in the decoder and the disassembler.
- 3 Execute. The two instructions in `mipster`, using C*'s own `<<` and `>>` — which do not exist in C* until your compiler from last week exists. Self-reference, as an assignment.
- 4 Generate. In last week's `compile_X`, emit them. Then `./grader/self.py bitwise-shift-execution` and `self-compile`.

RULES

Only `riscu.md` and `selfie.c`. No warnings. Note the bootstrap: to use shifts in `selfie.c`, `selfie` must first be compiled by a C compiler that has them, and then by itself.

Listen, then read.

LISTEN

Bach · Chaconne, from Partita No. 2 — Hilary Hahn.
One four-bar bass and sixty-four variations built over it: every expression a term over the same operand, evaluated in order, in fifteen minutes.

READ

The C Programming Language — Kernighan and Ritchie, 1978: the language C* is a subset of, in the book that taught it to everyone; the precedence table is on page 53. And, more technical, ALGOL 60 Translation — Dijkstra, 1961: the shunting-yard algorithm, and the first compiler for a language with recursive expressions.